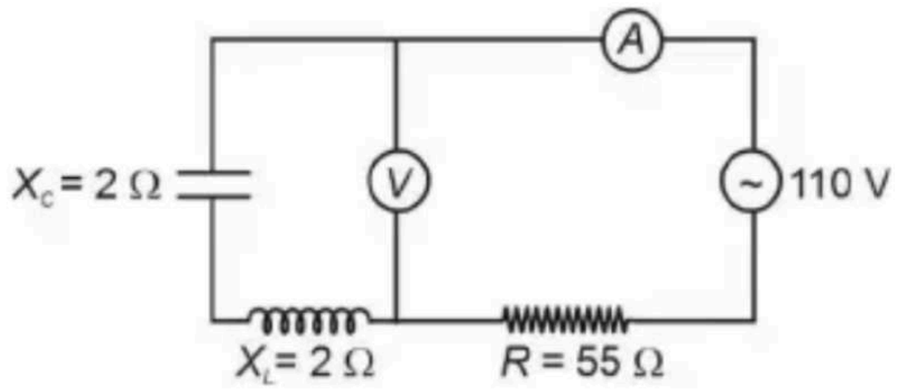
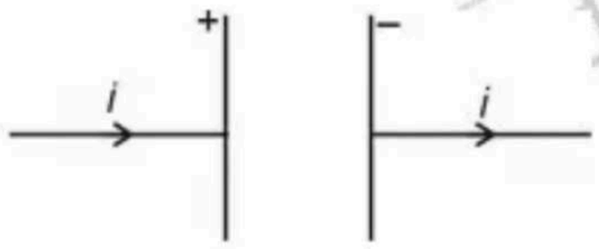
√

√


A
X_c = 2 Ω
V
~ 110 V
X_L = 2 Ω
R = 55 Ω

–

–


+
i
–
i

–

–

–

–

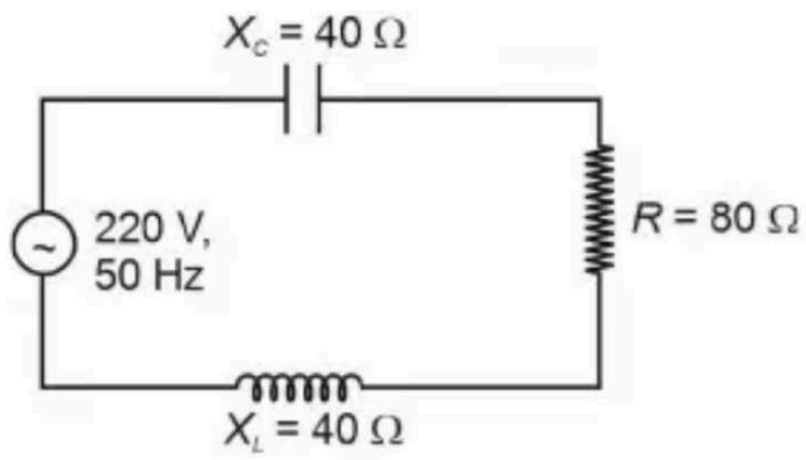
–

–

–

——

Aakash
Medical | IIT-JEE | Foundations



CHEMISTRY

54. Number of mole(s) of dichromate ion required to oxidize 0.3 mol of hydrogen sulphide in acidic medium will be

(1) 0.05 (2) 0.2

(3) 0.1 (4) 0.3

55. Incorrect statement among the following is

(1) Ce(IV) is more oxidizing in nature than water

(2) Eu(II) can act as reducing agent

(3) Yb(II) is acting as oxidant

(4) Nd may exhibit +4 oxidation state in its oxides

56. Correct electronic configuration of Sm (Samarium, Atomic number 62) is

(1) [Xe] $4f^6\ 6s^2$

(2) [Xe] $4f^5\ 5d^1\ 6s^2$

(3) [Xe] $4f^7\ 6s^1$

(4) [Xe] $4f^7\ 6s^2$

57. Correct order of ionic size among the following is (Atomic numbers : Eu = 63, Gd = 64, Er = 68, Tm = 69)

(1) $Eu^{3+} < Gd^{3+} < Er^{3+} < Tm^{3+}$

(2) $Eu^{3+} < Er^{3+} < Tm^{3+} < Gd^{3+}$

(3) $Tm^{3+} < Er^{3+} < Gd^{3+} < Eu^{3+}$

(4) $Tm^{3+} < Gd^{3+} < Er^{3+} < Eu^{3+}$

58. Actinoid shows only +3 oxidation state among the following is (Atomic numbers : Pa = 91, Am = 95, Cm = 96 and No = 102)

(1) Pa (2) Am

(3) Cm (4) Ac

59. Incorrect characteristics of actinoids is

(1) In finely divided condition, the actinoids are highly reactive in nature

(2) On reaction with boiling water, they give mixture of oxide and hydride

(3) 5*f* electrons of these elements are weakly shielded from the nuclear charge than 4*f* electrons of Lanthanoids

(4) Due to formation of protective oxide layers on its surface most of them slightly affected by nitric acid

60. Element of least melting point among the following is

(1) W

(2) Ta

(3) Ir

(4) Au

61. Neutral bidented ligand is

(1) ox (2) en

(3) gly (4) EDTA

62. Ligand of highest field strength as per spectro-chemical series is

(1) OH^- (2) CN^-

(3) H_2O (4) $C_2O_4^{2-}$

63. Electronic configuration of low spin octahedral complex in d^5 condition is

(1) $t_{2g}^3\ e_g^2$ (2) $t_{2g}^5\ e_g^0$

(3) $t_{2g}^2\ e_g^3$ (4) $t_{2g}^4\ e_g^1$

64. Number of rings present in complex $[Co(en)_2(ox)]Cl$ is

(1) 3 (2) 4

(3) 6 (4) 5

65. Which one of the following is expected to exhibit optical isomerism?

(1) Mer-$[Co(NH_3)_3Br_3]$

(2) Trans-$[Co(NH_3)_4Cl_2]Cl$

(3) $[Co(en)_3]^{3+}$

(4) Trans-$[CoCl_2(en)_2]$

66. Total number of unpaired electron(s) present in complex $[Ni(NH_3)_6]Cl_2$ is

(1) Zero (2) 2

(3) 3 (4) 4

67. Crystal field stabilization energy for low spin d^6 octahedral complex is (Δ_0 : Splitting energy gap, P : Pairing energy)

(1) $-2.4\Delta_0$

(2) $-2.4\Delta_0 + P$

(3) $-2.4\Delta_0 + 3P$

(4) $-1.2\Delta_0 + 2P$

68. Among the following in which metal carbonyls 'C–O' bond order is maximum?

(1) $[Cr(CO)_6]$ (2) $[Mn(CO)_6]^+$

(3) $[Fe(CO)_5]$ (4) $[V(CO)_6]^-$

69. Which one of the following complexes will exhibit the maximum paramagnetic behaviour?

(1) $K_2[NiCl_4]$ (2) $K_3[CoCl_6]$

(3) $Na_3[FeCl_6]$ (4) $K_4[ZnCl_6]$

70. Value of primary and secondary valencies in complex $[Zn(H_2O)(NH_3)(en)(gly)]Cl$ respectively are

(1) 2, 2 (2) 4, 4

(3) 3, 6 (4) 2, 6

71. Which of the following is a π-acid ligand?

(1) NH_2^- (2) OH^-

(3) CO (4) H_2O

72. Which complex is homoleptic?

(1) $[Cr(H_2O)_4Cl_2]Cl$

(2) $K_3[Fe(C_2O_4)_3]$

(3) $[CrCl_2Br_2(H_2O)_2]I_2$

(4) $[Co(en)_2ClBr]I$

73. EAN (Effective atomic number) of central metal atom in $[Fe(H_2O)_5NO]SO_4$ complex is

(1) 35

(2) 34

(3) 36

(4) 37

74. Aqueous solution of which of the given solution show highest conductivity?

(1) $CoCl_3.4NH_3$

(2) $CoCl_3.3NH_3$

(3) $CoCl_3.5NH_3$

(4) $CoCl_3.6NH_3$

75. Select the complex of highest stability among the following.

(1) $[Co(NH_3)_6]Cl_3$

(2) $[Co(en)_2\ Cl_2]Br$

(3) $[Co(H_2O)_6]Cl_3$

(4) $[Co(en)_3]Cl_3$

76. Correct increasing order of nucleophilicity in polar aprotic medium is

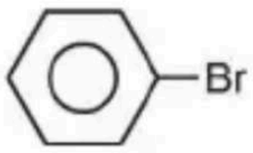
(1) $F^- < Cl^- < Br^- < I^-$

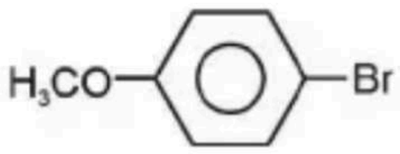
(2) $F^- < I^- < Br^- < Cl^-$

(3) $I^- < Br^- < Cl^- < F^-$

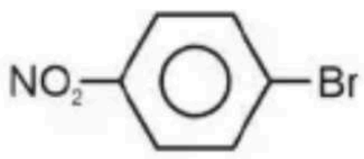
(4) $I^- < Cl^- < F^- < Br^-$

77. Which of the following compounds undergoes nucleophilic substitution reaction most easily?

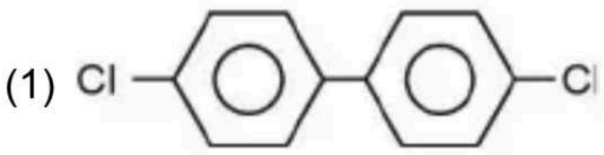
(1) 

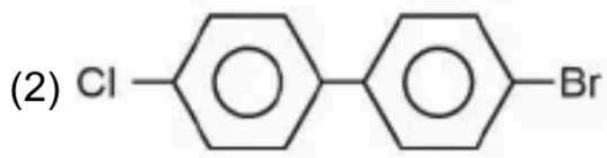
(2) 

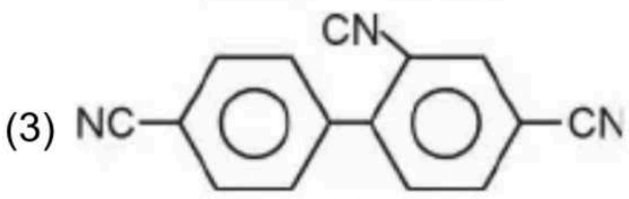
(3) 

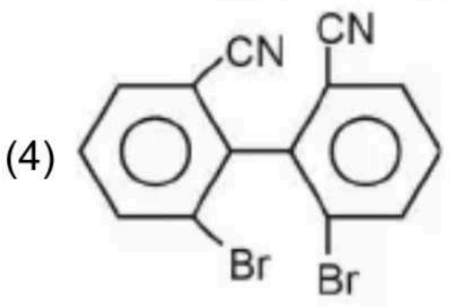
(4) 

78. Which of the following diphenyl is optically active?

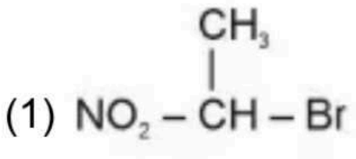
(1) 

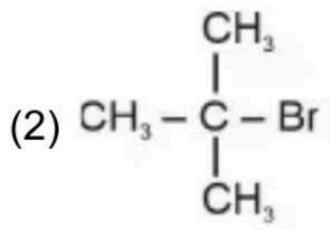
(2) 

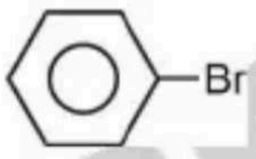
(3) 

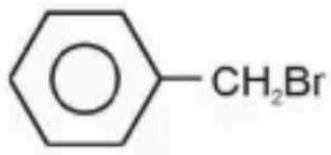
(4) 

79. In which of the following compounds, the C–Br bond ionization will give most stable carbonium ion?

(1) 

(2) 

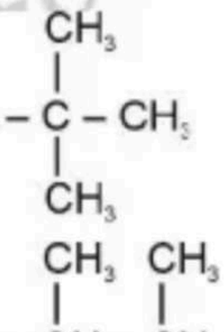
(3) 

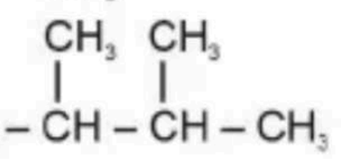
(4) 

80. The alkane that gives only one mono chlorinated product on chlorination with Cl_2 in the presence of UV light is

(1) $CH_3CH_2CH_3$

(2) $CH_3CH(CH_3)_2$

(3) 

(4) 

81. Select the incorrect statement among the following.

(1) Fluoro derivatives of hydrocarbons generally having more density than water

(2) The haloalkanes are very slightly soluble in water

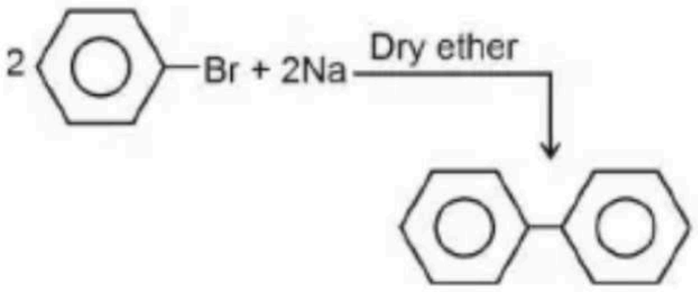
(3) Methyl bromide is in gaseous form at room temperature

(4) The boiling point of isomeric haloalkanes decrease with increase in branching

82. The correct order of E1 for alkyl halides is

(1) 1° > 2° > 3° (2) 3° > 2° > 1°

(3) 2° > 1° > 3° (4) 2° > 3° > 1°

83. 

This reaction is named as

(1) Fittig reaction

(2) Wurtz-fittig reaction

(3) Gattermann reaction

(4) Sandmeyer reaction

84. $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-CH=CH_2+HBr\rightarrow A(Major)$

Major product 'A' is

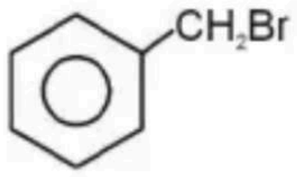
(1) $(CH_3)_3CCH_2CH_2–Br$

(2) $(CH_3)_3C-\underset{\displaystyle Br}{\overset{|}{CH}}-CH_3$

(3) $(CH_3)_2\underset{\displaystyle Br}{\overset{|}{C}}-CH(CH_3)_2$

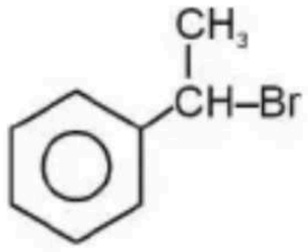
(4) $(CH_3)_2CH–CH(CH_3)_2$

85. Which of the following compounds will undergo racemisation when hydrolyses with solution of KOH?

(1) 

(2) CH_3Br

(3) $(CH_3)_2CH-CH_2Br$

(4) 

86. Which one give fastest rate of S_N2?

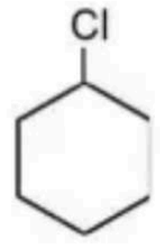
(1) CH_3Cl

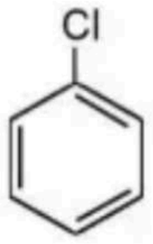
(2) $(CH_3)_2CH–Cl$

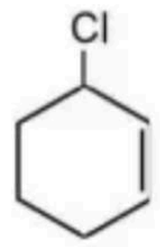
(3) $(CH_3)_3C–Cl$

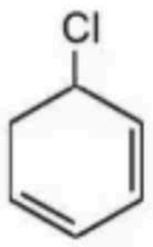
(4) $(CH_3)_2CH-\underset{\displaystyle Cl}{\overset{|}{CH}}-CH_3$

87. Which one easily give elimination on heating with aqueous solution of sodium ethoxide?

(1) 

(2) 

(3) 

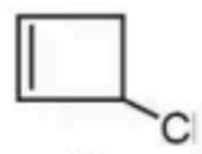
(4) 

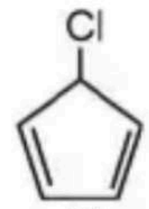
88. Among the following which methyl halide is of maximum dipole moment?

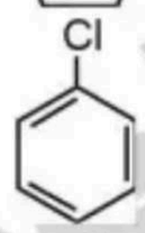
(1) CH_3F (2) CH_3Cl

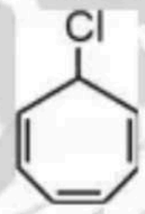
(3) CH_3Br (4) CH_3I

89. Which one gives fastest reaction with $AgNO_3$ solution?

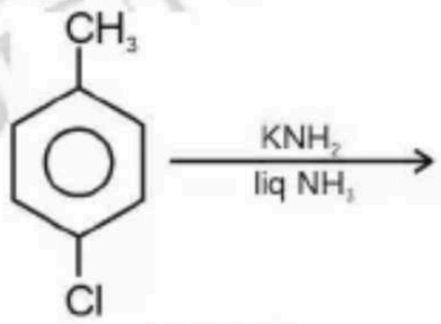
(1) 

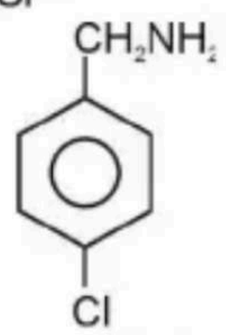
(2) 

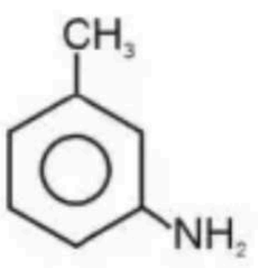
(3) 

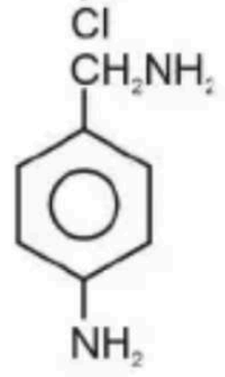
(4) 

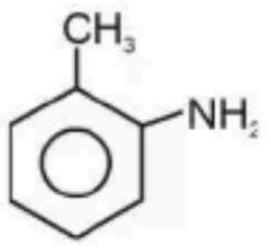
90. Major product of the following reaction is



(1) 

(2) 

(3) 

(4) 

BOTANY

91. A nitrogenous base is linked to the pentose sugar through a N-glycosidic linkage to form a
(1) Nucleotide
(2) Nucleolus
(3) Nucleus
(4) Nucleoside

92. Select the **mismatched** pair w.r.t DNA content

(1)	φ × 174	–	5386 nucleotides
(2)	Bacteriophage lambda	–	48502 base pairs
(3)	*E.coli*	–	2.5×10^3 bp
(4)	Human DNA (Haploid content)	–	3.3×10^9 bp

93. The acidic substance present in nucleus first identified by Friedrich Miescher in 1869 was
(1) RNA
(2) DNA
(3) H_1 protein
(4) Histone octamer

94. Which of the following is **not** common in both RNA and DNA?
(1) Cytosine
(2) Adenine
(3) Guanine
(4) 5-methyl uracil

95. Unequivocal proof that DNA is the genetic material came from the experiment of
(1) Walter Sutton
(2) Alfred Hershey and Martha Chase
(3) Thomas Hunt Morgan
(4) Frederick Griffith

96. A molecule which performs dynamic functions of messenger and adapter and is also found in some viruses as main genetic material has
(1) Deoxyribose sugar
(2) Thymine
(3) 2′OH group in their sugar
(4) More stable structure and shows no mutation

97. A molecule that can act as a genetic material should be able to fulfill all the following criteria **except**
(1) Generate its replica
(2) Chemically and structurally stable
(3) Show no changes or mutation at all
(4) Express itself in the form of 'Mendelian Characters'

98. Which of the following enzyme is responsible for the DNA synthesis from RNA?
(1) Ligase
(2) Reverse transcriptase
(3) RNA polymerase
(4) DNA polymerase I

99. Guanine is 20% in a double stranded RNA molecule. What would be the percentage of thymine in it?
(1) 30%
(2) 0%
(3) 60%
(4) 40%

100. Read the following statements and select the **correct** option.

Statement A : After the completion of replication, each DNA molecule would have one parental and one newly synthesised strand.

Statement B : DNA replicates semi conservatively in *E.coli* and conservatively in human.

(1) Both statements A and B are correct
(2) Only statement A is correct
(3) Only statement B is correct
(4) Both the statements A and B are incorrect

101. Average rate of DNA polymerization in *E. coli* is approximately
(1) 500 bp per second
(2) 100 bp per minute
(3) 1000 bp per minute
(4) 2000 bp per second

102. Select the **mismatched** pair w.r.t DNA replication.

(1)	dNTPs	–	Provides the energy for polymerization
(2)	DNA ligase	–	Join the discontinuous fragments of lagging strand
(3)	DNA replication	–	Occurs in S phase of cell cycle in eukaryotes
(4)	DNA polymerase-I	–	RNA dependent-DNA polymerase

103. In prokaryotes, unwinding of DNA double helix is done by
(1) DNA gyrase
(2) DNA polymerase I
(3) Helicase
(4) DNA ligase

104. Catalytic RNA in prokaryotes is
(1) 5 S rRNA
(2) 23S rRNA
(3) 18S rRNA
(4) 28S rRNA

105. In eukaryotes, 5S rRNA and SnRNA are synthesized by
(1) RNA polymerase I
(2) RNA polymerase II
(3) Topoisomerase
(4) RNA polymerase III

106. Rho factor is a protein in bacteria which is responsible for
(1) Initiation of DNA replication
(2) Termination of transcription
(3) DNA synthesis
(4) Protein synthesis

107. Polynucleotide phosphorylase enzyme
(1) Catalyse the DNA synthesis
(2) Is helpful in polymerising RNA with defined sequences in a template independent manner
(3) Add the phosphate to DNA at 5′ end
(4) Add the phosphate to DNA at 3′ end

108. All reference point while defining a transcription unit is made with
(1) Coding strand (2) Non-coding strand
(3) Non-sense strand (4) Template strand

109. Poly A tail present at the 3′ end of transcript is
(1) Composed of 200-300 adenylate residues
(2) Synthesized in template dependent manner
(3) Catalyzed by guanylyl transferase
(4) Mainly seen in prokaryotes

110. Mark the **incorrect** statement about transcription in prokaryotes
(1) Transcript is polycistronic
(2) Exons and introns are present in transcript of all prokaryotes
(3) Only one type of RNA polymerase synthesise all three types of RNA
(4) Sigma factor is an initiation factor

111. Select the **mismatched** pair w.r.t genetic code
(1) UAA – Stop codon
(2) AUG –Start codon in prokaryotes
(3) UGA –Stop codon
(4) AUU–Start codon in eukaryotes

112. Which of the following is **not** true for translation in prokaryotes?
(1) When the small subunit encounters an mRNA, the process of translation of the mRNA to protein begins
(2) There are four sites in large subunit of ribosome for incoming tRNA
(3) Ribozyme catalyzes the formation of peptide bond
(4) UTRs are required for efficient translation

113. How many amino acids will be there in peptide translated from the following transcript?

5′ AUG AUU AUG UAG AAA UGA 3′

(1) 5 (2) 6

(3) 4 (4) 3

114. In *lac* operon, repressor protein

(1) Bind to the *lac* i site

(2) Bind to operator region of operon

(3) Positively regulate the lac operon

(4) Bind to lac z gene

115. Which of the following gene of *lac* operon codes for β-galactosidase enzyme?

(1) *Lac i* (2) *Lac* y

(3) *Lac* z (4) *Lac* a

116. The actual structure of tRNA is a compact molecule which looks like

(1) A clover leaf

(2) A spiral staircase

(3) Linear straight line

(4) Inverted L

117. In prokaryotes the predominant site for the control of gene expression is at

(1) Regulation of splicing

(2) Transport of mRNA from nucleus to the cytoplasm

(3) Translational level

(4) Control of rate of transcriptional initiation

118. In a typical nucleus, euchromatin region is said to be

(1) Densely packed

(2) Transcriptionally active

(3) Darkly stained

(4) Transcriptionally inactive

119. Which of the following is **not** true about the HGP?

(1) The HGP was a 13-year project

(2) It was coordinated by U.S. Department of Energy and the National Institute of Health

(3) Project was completed in 1996

(4) ESTs was one of the approach which was used for sequencing in HGP

120. The technique of DNA fingerprinting developed by Alec Jeffreys, as used earlier, involved southern blot hybridisation using

(1) RT-PCR as probe

(2) Radiolabelled VNTR as probe

(3) Macro-satellite DNA

(4) NH_4Cl as the only nitrogen source

121. Chromosome 1 of human has _A_ genes and the Y has the _B_ genes respectively.

Fill in the blanks for A and B

(1) 2968, 231

(2) 1000, 200

(3) 230, 3000

(4) 200, 1000

122. Which of the following is a variety of wheat resistant to leaf and stripe rust?

(1) Pusa Komal

(2) Himgiri

(3) Pusa Sadabahar

(4) Pusa Shubra

123. Changes in the base sequence within genes through use of chemicals or radiations and selecting and using the plants that have the desirable characters as a source of breeding, is called

(1) Conventional breeding

(2) Biofortification

(3) Mutation breeding

(4) Germplasm collection

124. Pusa Swarnim is a variety of

(1) *Brassica* resistant to white rust

(2) Cowpea resistant to bacterial blight

(3) Chilli resistant to black rot

(4) Wheat resistant to hill bunt

125. Parbhani kranti variety of *A. esculentus* is resistant to

(1) Yellow mosaic virus

(2) Tobacco mosaic virus

(3) Leaf curl

(4) Curl blight black rot

126. Select the **incorrectly** matched pair w.r.t varieties of crop
(1) Pusa sem 2 – Flat bean
(2) Pusa A-4 – Okra
(3) Pusa Gaurav – *Brassica*
(4) Pusa Komal – Wheat

127. Atlas 66, having high protein content, is a variety of
(1) Sunflower (2) Wheat
(3) French bean (4) Rice

128. International Rice Research Institute is located at
(1) New Delhi
(2) Kolkata
(3) Mexico
(4) Philippines

129. Taichung Native -1 is
(1) A dwarf rice variety
(2) A semi-dwarf variety of rice
(3) Developed in IARI, India
(4) Dwarf variety of wheat

130. Select the **odd** one w.r.t *Saccharum officinarum*
(1) Thick stem
(2) High sugar content
(3) Did not grow well in north India
(4) Has very poor yield

131. In mung bean, resistance to __A__ and powdery mildew are induced by mutations.
Fill in the blank with **correct** option
(1) Tobacco Mosaic virus
(2) Aphids
(3) Yellow mosaic virus
(4) White rust

132. Pomato a hybrid of potato and tomato was developed through
(1) Conventional breeding methods
(2) Somatic hybridization
(3) Induced mutation
(4) Meristem culture

133. The capacity to generate a whole plant from any cell is called
(1) Cybrid (2) Somatic hybrid
(3) Totipotency (4) Somaclone

134. Select the **incorrect** statement about plant tissue culture
(1) Growth regulators like auxins, cytokinins are used
(2) Sucrose can be used as carbon source
(3) Vitamins and amino acids are used in nutrient medium
(4) Meristem culture is not free of virus

135. Breeding crops with higher levels of vitamins and minerals, or higher protein content and healthier fats is called
(1) Biomagnification
(2) Bioaccumulation
(3) Biofortification
(4) Bioconcentration

ZOOLOGY

136. Urey and Miller experimentally proved
(1) Biogenic evolution of organic substances
(2) Abiogenic evolution of organic substances
(3) Theory of natural selection
(4) Theory of biogenesis

137. The earliest cells during reducing environment were
(a) Prokaryotes
(b) Thermophilic
(c) Chemoheterotrophs
(d) Anaerobes
(e) Thermolabile
(f) Chemoautotrophs
(1) (a), (e), (f) only
(2) (a), (b), (f) only
(3) (a), (d), (e), (f) only
(4) (a), (b), (c), (d) only

138. Choose the content in spark discharge apparatus used by Miller and Urey in their experiment

	Content	**Ratio**
(1)	$CH_4 : H_2 : NH_3$	2 : 1 : 2
(2)	$H_2O : CH_4 : NH_3$	1 : 2 : 1
(3)	$CH_4 : NH_3 : H_2$	2 : 1 : 2
(4)	$NH_3 : CH_4 : H_2$	1 : 1 : 2

139. Which organic polymer act as a catalyst in origin of life?

(1) DNA (2) RNA

(3) Protein (4) Enzyme

140. Choose the **incorrect** statement

(1) Coacervates do not fulfil the requirements for being precursor of life

(2) During chemical evolution, there was an accumulation of organic molecules in water

(3) Theory of biogenesis could not explain origin of life

(4) Life could not originate nowadays because of presence of reducing atmosphere and microbes

141. Excess use of pesticides or herbicides resulted in selection of resistant varieties in a much lesser time scale. It is an example of

(a) Evolution by anthropogenic action

(b) Diversifying selection

(c) Evolution is not a direct process in the sense of determinism but is a stochastic process based on chance events in nature and chance mutations in the organisms.

(1) (b) only (2) (a) only

(3) (a), (b) only (4) (a), (c) only

142. In humans, the two sets of muscles that move the ears are

(1) Analogous structures

(2) Homologous structures

(3) Vestigial structure

(4) Phylogenetic structures

143. Complete the analogy

Unit of evolution : Population : : Unit of natural selection : _____

(1) Individual (2) Species

(3) Family (4) Population

144. "Some times change in allele frequency is so different in the new sample of population that they become a different species" which of the term given below describe this scenario?

(1) Founder's effect (2) Speciation

(3) Saltation (4) Divergence

145. Animals like bandicoot, wombat, banded anteater, sugar glider are examples of

(a) Adaptive radiation

(b) Convergent evolution

(c) Marsupials of Australia

(d) Metatherian mammals

(1) (a), (c) only (2) (b), (c)

(3) (a), (c), (d) (4) (b), (c), (d)

146. Which of the following period is known as age of cycads?

(1) Triassic (2) Jurassic

(3) Permian (4) Cretaceous

147. All of the following are examples of homologous organs **except**

(1) Thorns of *Bougainvillea* and tendrils of *Passiflora*

(2) Brain of fish, frog and cat

(3) Legs of insects

(4) Flippers of penguin and dolphins

148. *Triceratops* and *Stegosaurus* were

(1) Extant reptiles

(2) Dinosaurs

(3) Extinct birds

(4) Extant birds

149. The geological time scale is divided into eras, periods and epochs. Under which of the following era Silurian period falls?

(1) Proterozoic

(2) Palaeozoic

(3) Mesozoic

(4) Coenozoic

150. The process of evolution of different species in a given geographical area starting from a point and literally radiating to other areas of geography is called
(1) Coevolution
(2) Adaptive radiation
(3) Convergent evolution
(4) Parallel evolution

151. Which of the following organism is phylogenetically close living relative of birds?
(1) Dinosaurs (2) Mammals
(3) Crocodiles (4) Tuatara

152. The finches of Galapagos islands provide an evidence in favour of
(1) Embryological evidences
(2) Biogeographical evidences
(3) Paleontological evidences
(4) Special creation

153. What was the feeding habit of founder species of Darwin finches?
(1) Insectivorous (2) Piscivorous
(3) Cactus eater (4) Seed eater

154. Darwinian variations are
(1) Small and directional
(2) Random and directionless
(3) Small and directionless
(4) Random and directional

155. Frequency of the mutated gene in a population is expected to increase if that gene is
(1) Dominant
(2) Recessive
(3) Inheritable
(4) Adaptable and selected by nature

156. If a population of organism is introduced into a new territory where environmental conditions are favourable, the population grows rapidly because
(1) Rate of reproduction increases
(2) Predation is nil
(3) Food supply becomes limited
(4) More offsprings survive and adaptive radiation occurs

157. The ship on which Darwin was appointed as naturalist was
(1) Sea gull
(2) H.M.S. Beagle
(3) Century
(4) H.M.S. Biegle

158. Genetic drift
(1) Is an orderly change in gene frequencies
(2) Is a random change in gene frequencies
(3) Produce great fluctuations in large populations
(4) Has nothing in common with inbreeding

159. In our modern understanding of natural selection, the fittest individuals are those who are
(1) Best adapted to the environment
(2) Best equipped to cope with special environmental conditions
(3) More contributor to the gene pool
(4) Producing many offspring but only few survive

160. To be a successful event for evolution, mutation must occur in
(1) Somatoplasm DNA
(2) Plasma protein
(3) RNA
(4) Germplasm DNA

161. Which type of selection indicate that extreme values have the highest fitness and mean values are relatively disadvantageous?
(1) Directional selection
(2) Diversifying selection
(3) Progressive selection
(4) Normalising selection

162. Which of the following theory is **not** given by Darwin?
(1) Theory of pangenesis
(2) Artificial selection theory
(3) Natural selection theory
(4) Theory of recapitulation

163. Which of the following is/are key concept of Darwin's theory of evolution?

(a) Natural selection

(b) Use and disuse of organs

(c) Branching descent

(d) Reproductive isolation and speciation

(1) (a), (c)

(2) (b), (c)

(3) (a), (d)

(4) (c), (d)

164. Hardy Weinberg principle states that sum total of all allelic frequencies in a evolutionary population are

(1) Equivalent to 1

(2) More or less than 1

(3) $(p + q)^2 = 1$

(4) $p = 1 - q^2$

165. Select the **correct** option w.r.t. given statements

A : Bottle neck effect explains evolution of Darwin finches on Galapagos island

B: Genetic drift can bring change in gene frequency by chance in any direction

(1) Both A and B statements are correct

(2) Only A is correct

(3) Only B is correct

(4) Both A and B statements are incorrect

166. Which of the following is the **incorrect** match?

(1) A. Weissmann – Theory of germplasm

(2) Hugo de Vries – Mutation theory

(3) Thomas Malthus – An essay on principle of population

(4) Oparin – Theory of evolution of life

167. Select the nearest ancestor of mammals from the following representatives of evolutionary history of vertebrates

(1) Sauropsids (2) Synapsids

(3) Thecodonts (4) Therapsids

168. Around how many years back, agriculture settlement came into existence and leads to human settlement?

(1) 18,000 years ago

(2) 10,000 years ago

(3) 75,000 years ago

(4) 40,000 years ago

169. Which of the following human ancestor had cranial capacity as that of modern man?

(1) *Homo habilis*

(2) *Homo erectus*

(3) *Homo neanderthalensis*

(4) *Australopithecus*

170. Prehistoric cave art can be seen in Bhimbetka rock shelter in __X__ district of __Y__ state. Fill the blanks with **correct** options

	X	Y
(1)	Hoshangabad	Madhya Pradesh
(2)	Raisen	Madhya Pradesh
(3)	Hamirpur	Himachal Pradesh
(4)	Kurnool	Andhra Pradesh

171. Skull of baby chimpanzee is more like skull of

(1) Adult chimpanzee

(2) Human baby

(3) Adult human

(4) Neanderthal man

172. Which of the following statement is **incorrect**?

(1) About 15 mya, *Ramapithecus* and *Dryopithecus* were existing.

(2) *Ramapithecus* were more like man

(3) The first human like being the hominid was *Homo habilis*

(4) *Homo erectus* used animal hides to protect their body and buried their dead ones

173. Select the **odd** one w.r.t divergent evolution

(1) Types of beaks in Darwin finches

(2) Heart of amphibians and reptiles

(3) Eyes of *Octopus* and fish

(4) Forelimbs of whale and bat

174. Which of the given options does **not** support Darwinism?

(1) Mimicry

(2) Symbiosis

(3) Predator-prey relationship

(4) Vestigial organs

175. The basic process which will lead to disruption of Hardy Weinberg equilibrium is absence of

(1) Mutation

(2) Natural selection

(3) Random mating

(4) Gene flow

176. During ice age between _____ to _____ years ago, the modern day man arose. Fill the blank with **correct** option

(1) 1,00,000 – 40,000

(2) 75,000 – 10,000

(3) 18000 – 10,000

(4) 2000 – 2500 million

177. Choose the **incorrect** match linking human evolution to their features

(1)	*Australopithecus*	–	Essentially ate fruits
(2)	*Homo habilis*	–	Probably did not eat meat
(3)	*Homo neanderthalensis*	–	Lived in near east and central Asia
(4)	*Homo erectus*	–	Brain capacity were between 650-800 cc

178. Dinosaurs suddenly disappeared from the earth about

(1) 56 mya (2) 15 mya

(3) 16 mya (4) 65 mya

179. Numbat show similarity with

(1) Mole (2) Mouse

(3) Lemur (4) Anteater

180. Read the following statements-A and B, choose the **correct** option.

A: Pouched mammals of Australia show restricted distribution.

B: Due to continental drift and lack of competition from any other mammal, they survived in Australia.

(1) Both statement A and B are correct

(2) Both A and B statements are incorrect

(3) Only A statement is correct

(4) Only B statement is correct

❑ ❑ ❑